

AI Usage Report

Course: Software for Mobile Devices

Assignment: Assignment 1 - Flex++

Student Name: _____

Registration No.: _____

Date: _____

1. AI Tool(s) Used

ChatGPT and AI coding assistance through Pi/Codex were used for requirements discussion, implementation, debugging and independent review. The application itself contains no runtime AI service, model API, chatbot or API key.

2. Purpose of AI Usage

AI assistance supported a mobile academic-planning solution: reusable React Native components, conditional view switching, data-driven course lists, attendance recovery, weighted marks, GPA calculations, saved scenarios and linked tasks. It also supported chart-kit integration, input validation, storage handling, test generation, debugging and documentation. Synthetic data allowed meaningful interactions without accessing live FLEX records. The scope was reduced to a coherent academic workflow rather than replicating every portal page.

3. Important Prompts Used

The following are condensed summaries of important development instructions.

Prompt 1: Develop a React Native and JavaScript academic dashboard using conditional view switching, reusable components and two different react-native-chart-kit chart types. Use synthetic data and do not add runtime LLM usage.

Prompt 2: Make attendance and marks scenarios accurate and explainable. Preserve pending or unpublished records as unknown, save assumptions rather than results, and recalculate saved plans when baseline data changes.

Prompt 3: Review the implementation against the assignment and repair incorrect calculations, persistence status and scenario transfer. Retire unnecessary service, registration, outcomes and document features; verify the retained product and package only relevant source files.

4. AI-Generated Output

AI drafted substantial parts of the application code, tests and documentation. The output included the conditional app shell, shared UI components, synthetic records, attendance/marks/GPA functions, dashboard charts, validated task forms, saved-plan comparisons and AsyncStorage persistence. Initial output required correction; successful compilation and passing tests were not treated as proof that all workflows were correct.

5. Changes Made by Me

Changes adopted in the final solution were carried out through the AI-assisted development and review process. They included reducing peripheral portal features, keeping explicit grade assumptions separate from hypothetical marks, fixing scenario-conditioned targets and uncertainty, distinguishing impossible attendance schedules, preserving course inputs across section changes, and enabling transfer into named plans.

Further AI-assisted revisions corrected premature save-success messages, serialized/coalesced storage writes, added malformed-data recovery, removed hidden grade assumptions, separated linked-action state by course/plan, exposed task editing and added attendance-session selection.

6. My Understanding

App.js switches views with state and conditional rendering. Screens receive records and action callbacks through props. Arrays of courses, enrollments, assessments and sessions generate lists with map, while filter selects the relevant term or course. Shared components keep form feedback and presentation consistent.

Attendance uses present sessions divided by recorded held sessions. Pending entries remain provisional. For example, 7/10 becomes 12/15 after five attended sessions, reaching 80%; this is feasible only if enough sessions remain. Weighted marks use assessment weight multiplied by obtained marks divided by maximum marks. Unpublished results remain unresolved.

GPA uses credit-weighted grade points; F/FA retain their credits at zero points, while W/I/pending records are excluded. A saved plan stores inputs, so changed records produce new calculations. AsyncStorage persists local state; accepting an edit into memory is different from confirming that its disk write succeeded.

7. Verification and Testing

Automated checks were executed during AI-assisted development and independently rerun against the final source package. npm run verify passed 193 tests, 52 import-check files, 33 hook-check files, 43 Node screen renders, 29 persistence checks and 38 store-contract checks. Expo Doctor passed 21/21 checks, and Android bundle export succeeded. These results establish the stated automated checks, not complete native usability.

Review found errors including missing values displayed as zero, saved marks calculations ignoring hypothetical scores, infeasible attendance shown as successful, course inputs lost between sections, and save status reported before the latest write completed. The retained implementation was revised and regression-checked. Selected tests use actual React with native modules stubbed; storage checks exercise delayed and failed writes.

Phone execution through Expo Go via QR scanning was reported, and seven supplied Android screenshots were visually reviewed. They show the dashboard, attention cards, both chart types, module standings and shortcuts at different scroll positions. Review identified status-bar/header overlap and crowded attendance-chart labels. Course and task interactions, keyboard behavior, hardware Back and restart persistence remain unverified. No publication or grading-system submission is claimed.

8. Reflection

The main lesson is that AI can accelerate implementation but its output still needs explicit requirements, arithmetic checks and realistic workflow testing. Separating domain calculations from presentation makes errors easier to inspect. Reducing optional features improves coherence and leaves more attention for correctness and mobile usability. AI assistance is most useful when its suggestions are checked against evidence rather than accepted because the code compiles.

Student Declaration

I confirm that I have used AI tools only as a development assistant and that I understand the code submitted as part of this assignment. I am able to explain and demonstrate the functionality of my application.

Student Name: _____

Signature: _____

Date: _____